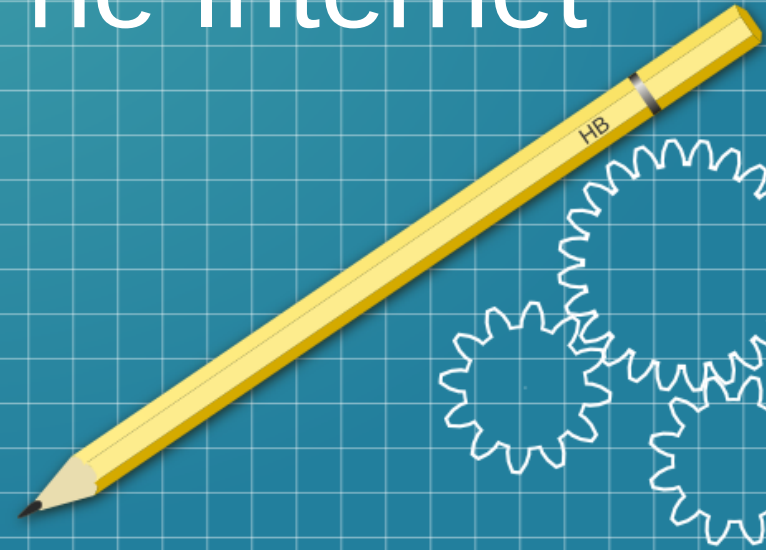
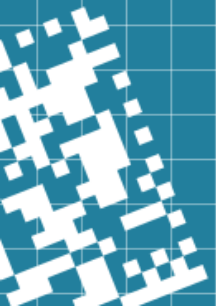
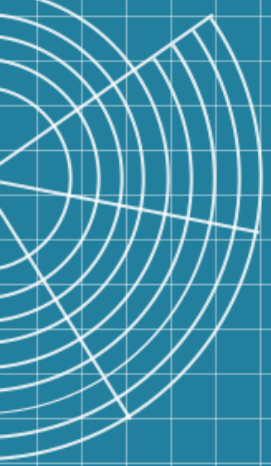
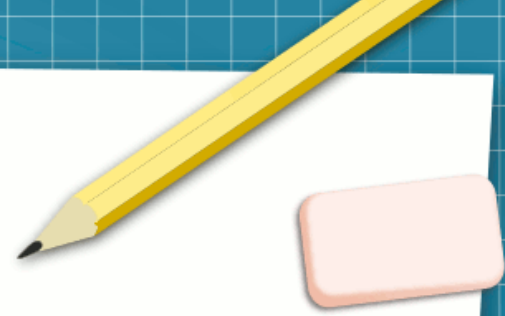


Computer Networks & The Internet



Hardware

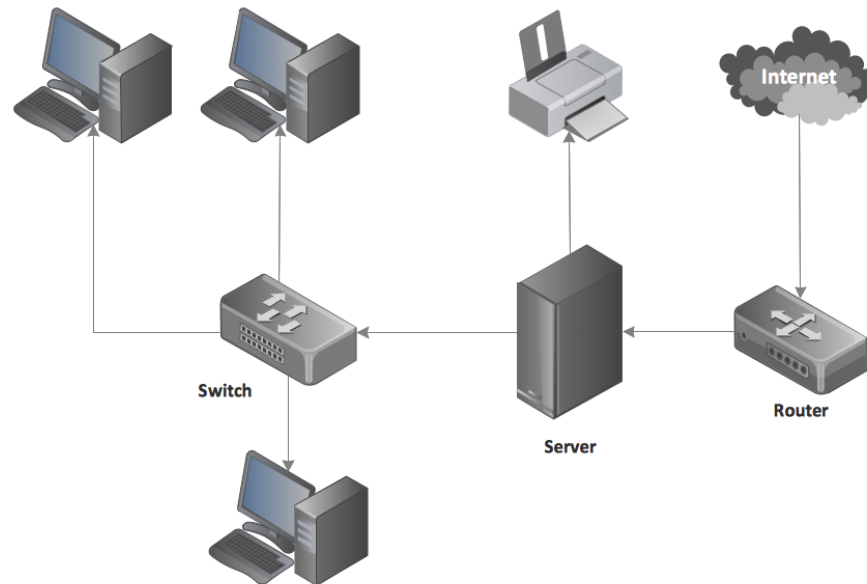
- Client (Servers / Laptops / Mobiles / PCs)
- Switches
- Routers
- WiFi Access Points (WAP)
- Cables (Fibre Optics , Copper)
- Satellites



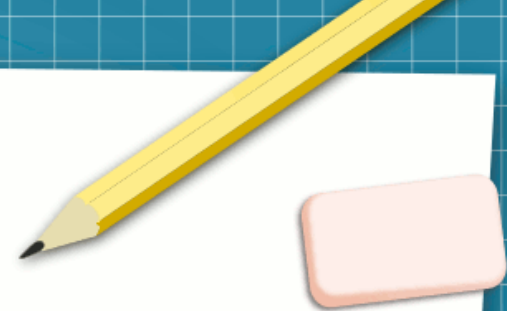
What is a Computer Network

- More than one Client connected together to share/pass data

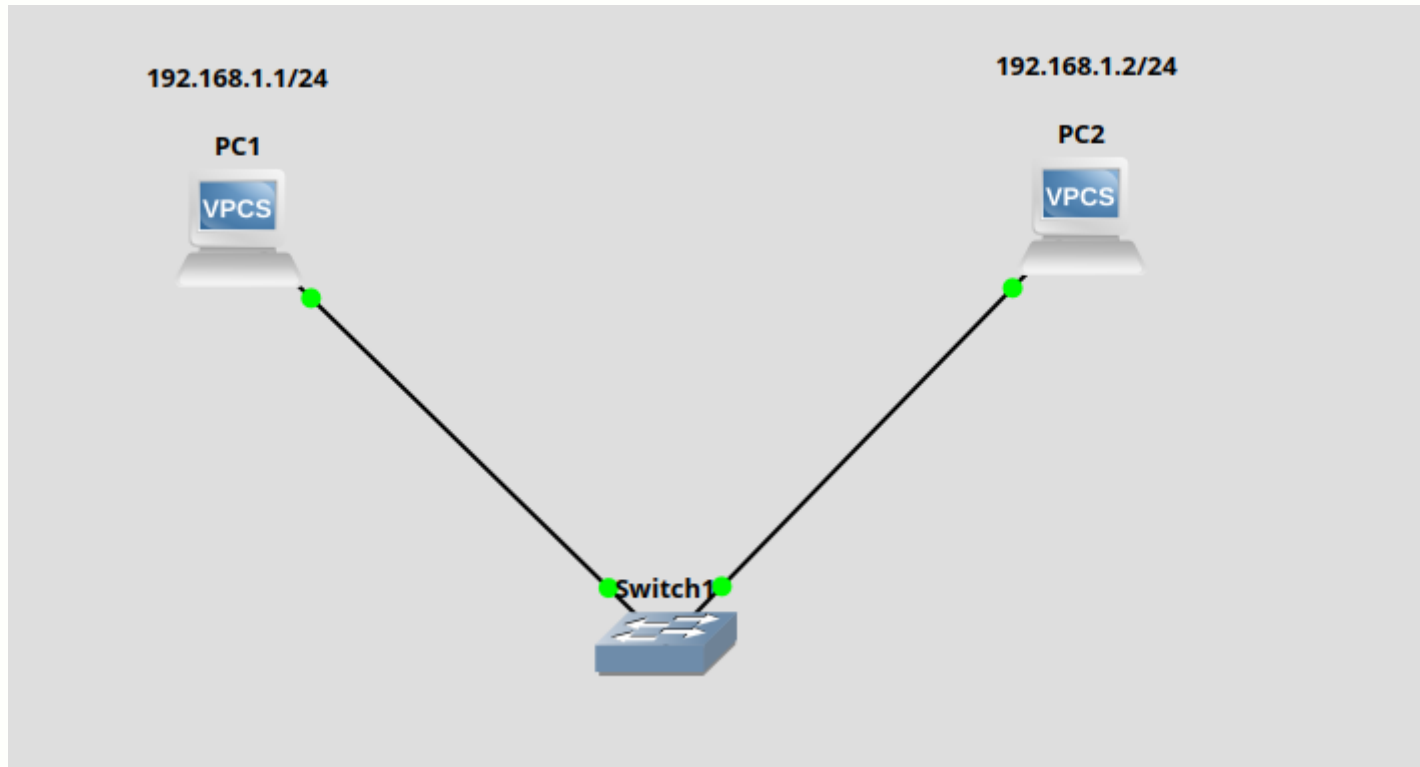
Sample Network Diagram



Point A to Point B

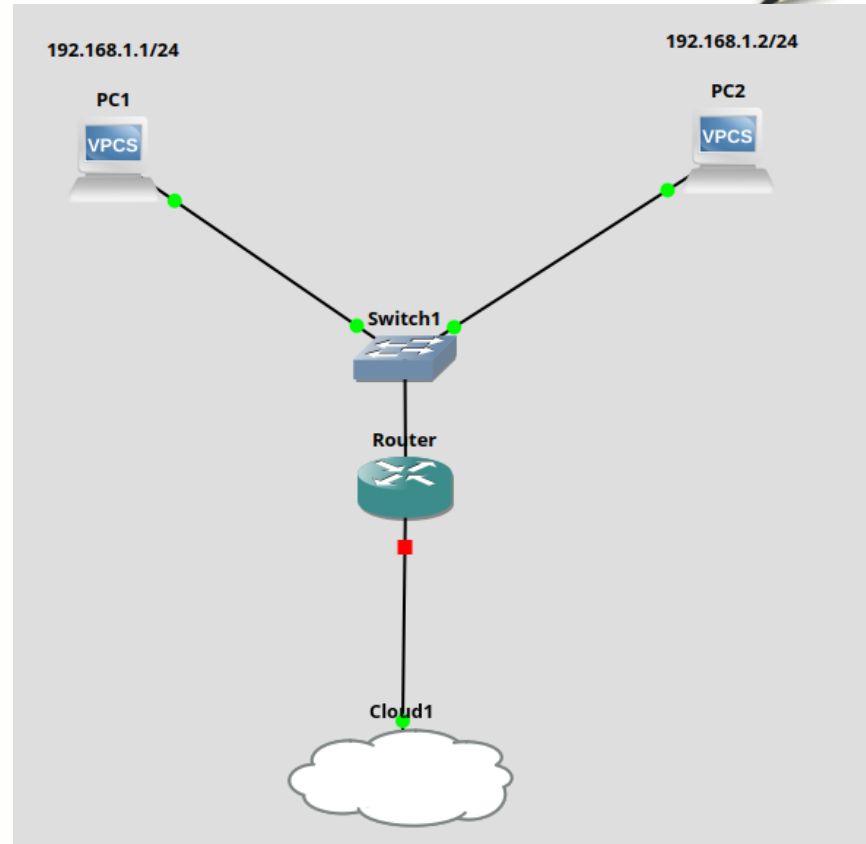


Switches

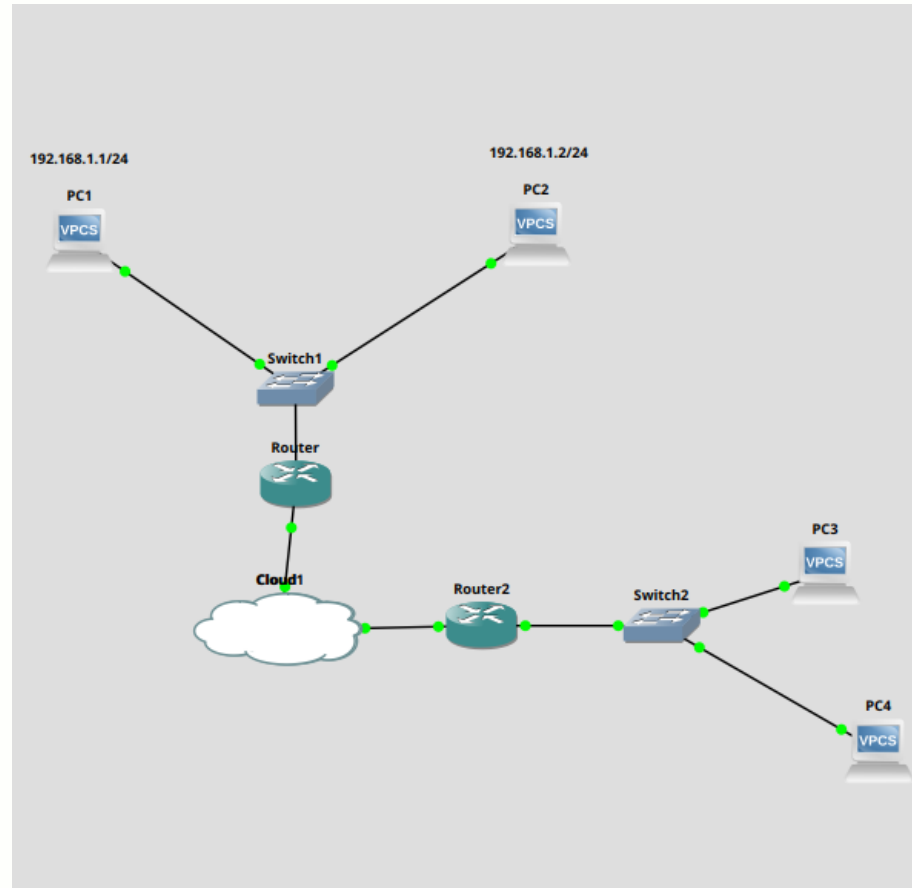


Routers

Typical Home network
but without WiFi



Networks of Networks



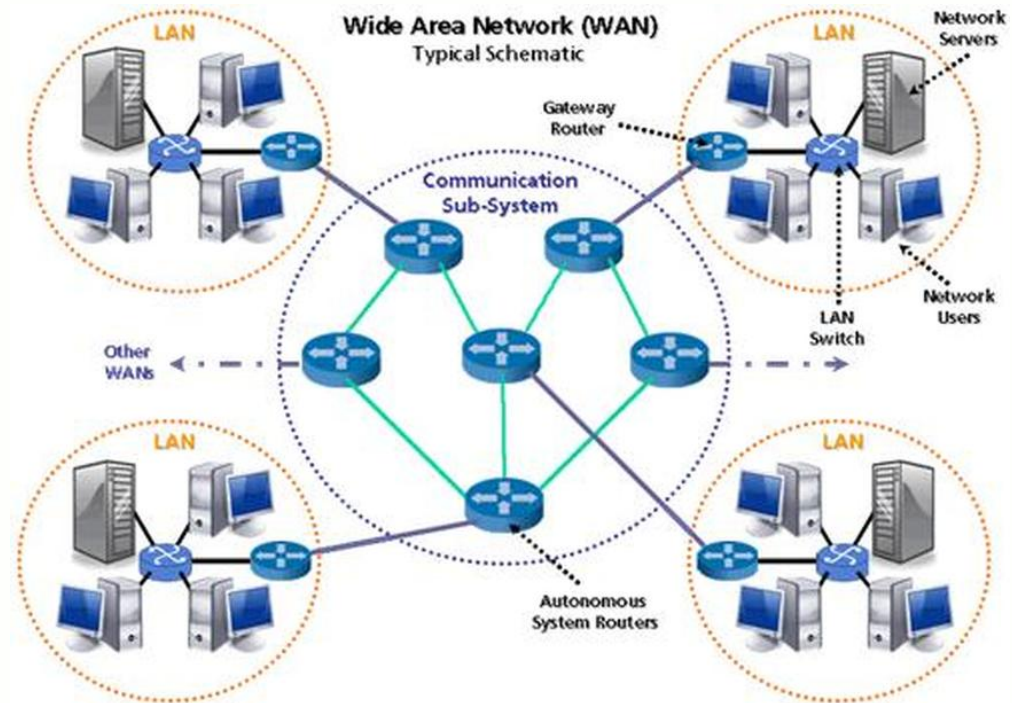
LAN / WAN

- Local Area Network (LAN)
 - Home
 - School
 - Offices
- Wide Area Network (WAN)
 - Large network spanning large geographic area
 - Internet is an extension of a WAN



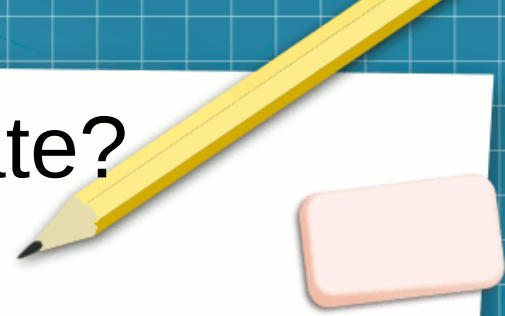
What is the Internet?

- A Network of Computer Networks

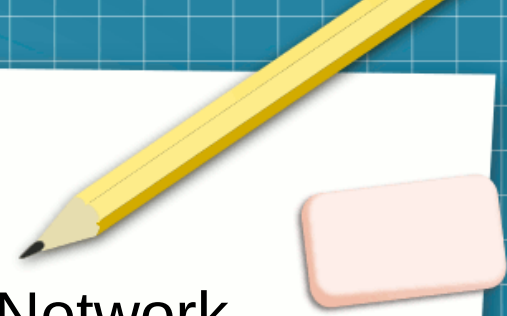


How do computers communicate?

- Computers have addresses like people do.
 - MAC address (Media Access Control address)
 - IP address (Internet Protocol Address)



MAC Address



- MAC address is a unique identifier assigned to a Network Interface controller (NIC)
- Format of a MAC address is a 48bit hex string split into 6 octets
 - eg aa:bb:cc:11:22:33
- With this there is over 281 trillion possible combinations of MAC address

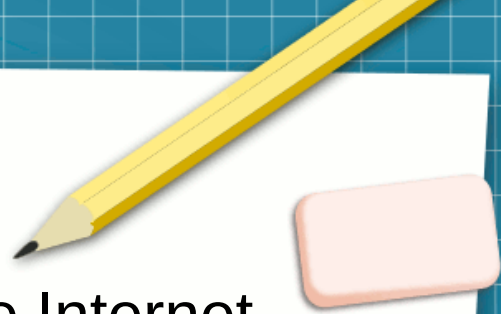
IP Address



- An IP address is an numerical address assigned to a computer connected to a computer network.
- IP addresses are 32 bit number split into 4 octets
 - eg 192.186.1.1
- There are about 4.2 Billion IPv4 addresses and these are used to communicate with computers over the internet
- There are Public and Private IP network address spaces
- There is also IPv6 which is not as commonly used but has 340,282,366,920,938,000,000,000,000,000,000,000* unique IP addresses

*340 undecillion

IP Addresses



- Local/Private IP addresses – Not routable over the Internet
 - 192.168.1.1
 - 10.0.1.1
 - 172.16.1.1
- WAN IP address – Example of some public IP addresses
 - 151.101.0.81
 - 1.1.1.1
 - 8.8.8.8

How Do I Reach Google.com



- Computers can't speak English only Binary so how can we get to google.com
- google.com resolves to IP addresses
 - 142.251.29.100 > Binary 10001110.11111011.00011101.01100100
 - 142.251.29.101 > Binary 10001110.11111011.00011101.01100101
 - 142.251.29.102 > Binary 10001110.11111011.00011101.01100110
 - 142.251.29.113 > Binary 10001110.11111011.00011101.01110001
 - 142.251.29.138 > Binary 10001110.11111011.00011101.10001010
 - 142.251.29.139 > Binary 10001110.11111011.00011101.10001011

DNS (Domain Name System)



- This is like the Phone book of the internet and translates domains to IP addresses
 - bbc.com – 151.101.0.81
 - google.com – 142.251.29.100
- DNS Servers – These are the DNS servers that give you the IP addresses. Here are some common ones.
 - 1.1.1.1 / 1.0.0.1 – cloudflare DNS
 - 8.8.8.8 / 8.8.4.4 – Google DNS
 - 9.9.9.9 - Quad9

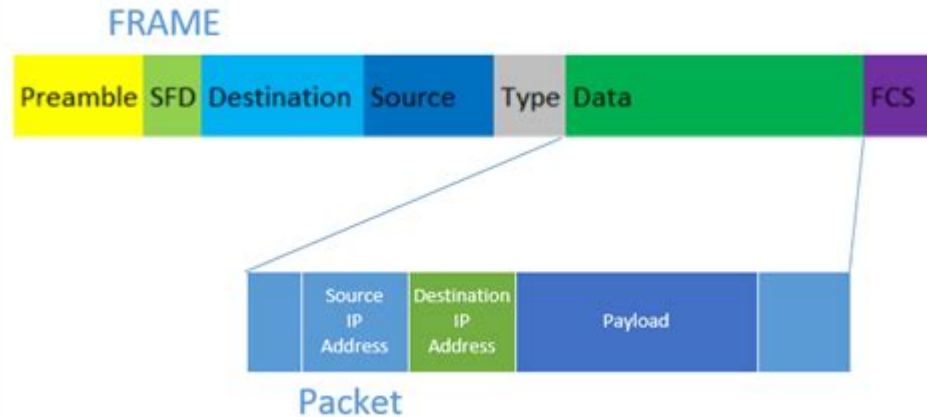
Now we know where we are going



- Next is what we are going to send.
- Going to google.com in your browser `Chrome, Firefox, Edge`
- This is a HTTP/S request and when requesting a website/page
 - The first request is a GET#
- We put this request inside of a packet that then gets sent down the wire (or over the air for WiFi) to your switch/router to then make its way to google.com IP address

Packets & Frames

- Data is sent over these Networks in Ethernet Frames, These contain packets.



IP Packet

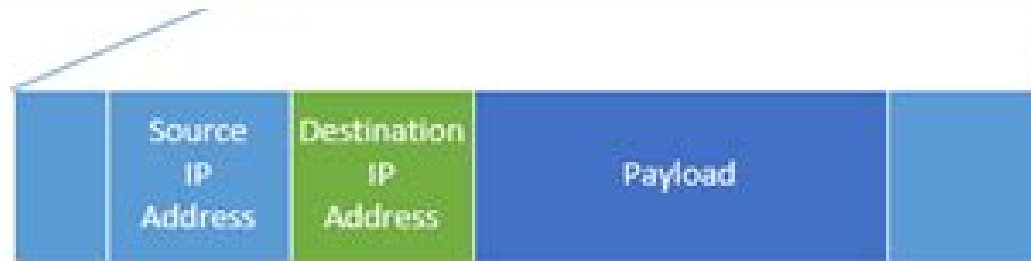


The IP Packet is part of the TCP/IP Protocol which is the main protocol of the Internet.

Source IP : Where is this packet coming from

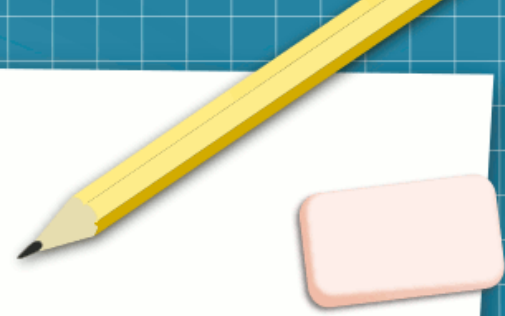
Destination IP: Where is this packet going

Payload: The data that is being sent



Packet

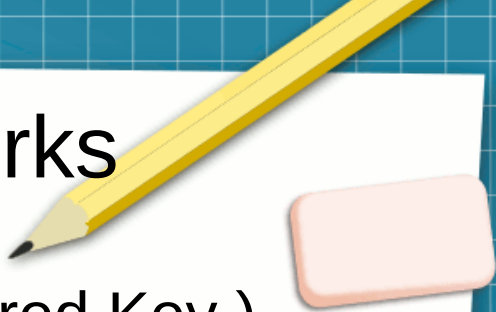
Cat Pic



How Does the Internet Work



Secure and Non Secure Networks



- Networks like WiFi that use a password (Pre Shared Key), depending on the security type are pretty secure.
- Public WiFi like at Coffee shops and Hotels that say `Free` WiFi, and you do not need a password are not secure.
- On Ethernet/Cable, these are pretty secure but can be wire-tapped.

So how do we secure our Internet Traffic?

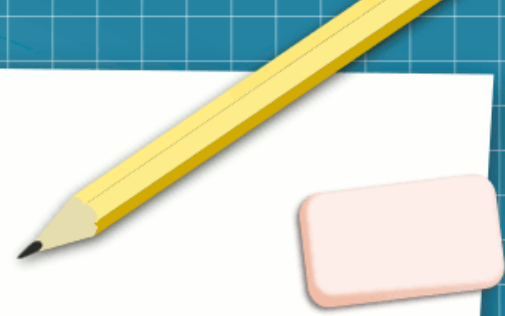
HTTPS



- Using protocols Like HTTPS which connects to the server to establish a secure encrypted communication to the server
- Plain HTTP shows all text being sent with no encryption.
- This is what is used on most sites these days so make sure you look for the green lock(on most browsers)



VPNs



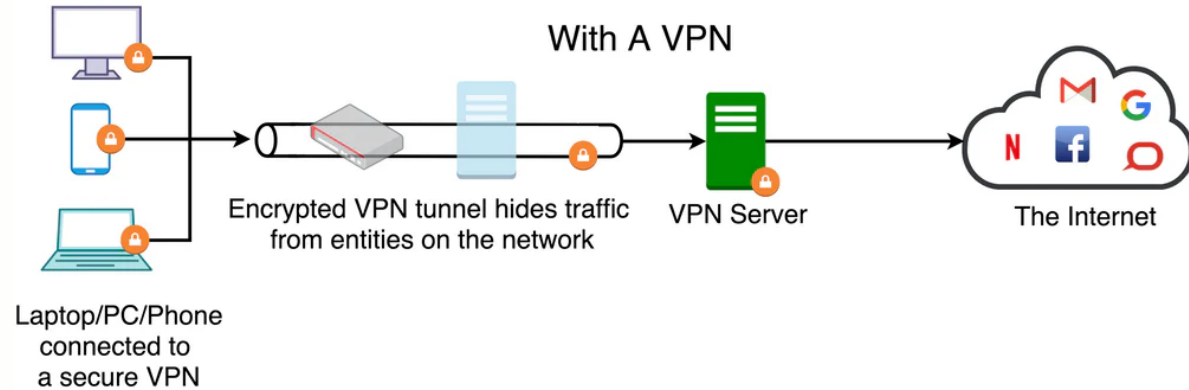
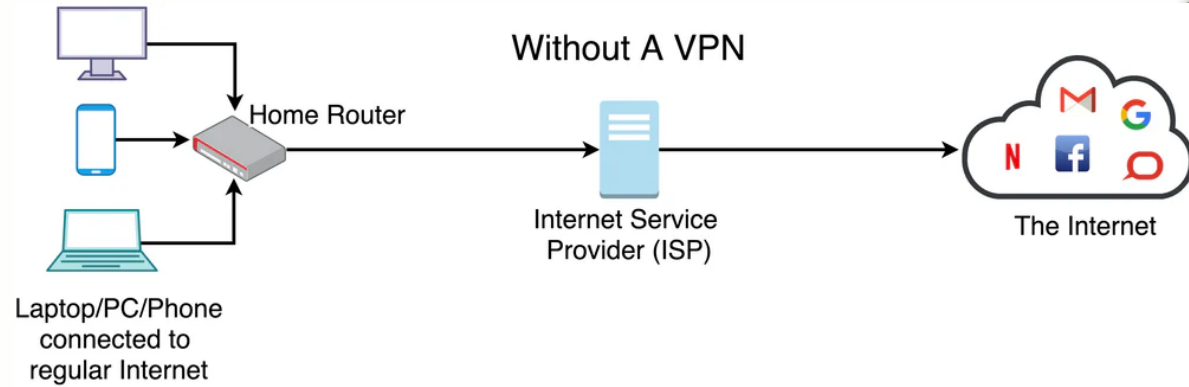
- Virtual Private Networks
 - Networks that connect two clients together with an encrypted tunnel
- VPNs are used by businesses so workers can Connect to their office servers
- VPNs are used to protect data on Public WiFi
- VPNs are used for secure communications
- VPNs can be used appear as if you are in a different part of the world.

VPNs

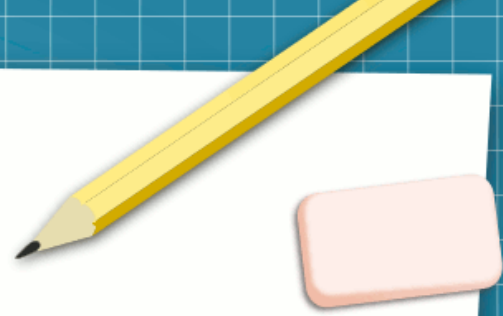
- VPNs get given a bad reputation as they are also used by Bad actors/Hackers/Criminals
- VPNs 'Can' be used to bypass country restricted content
 - eg. Being able to watch US Netflix from UK via a VPN that appears that you are in the US.
- VPNs are Essential for Privacy online.

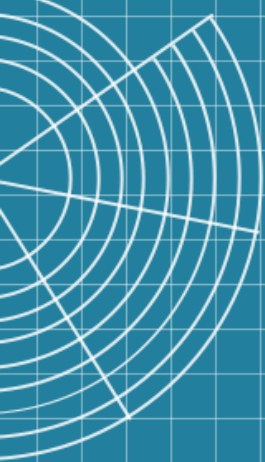


VPN Visual



Questions?





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